

INTERNSHIP EXPERIENCE – CHRONUS SOFTWARE INDIA

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ABOUT THE COMPANY:

Chronus Software India is a product-based company. It focuses on building mentoring software for other companies. Its office is located in Chennai.

ROLE DETAILS:

I was offered the role of intern for 2 months. The internship will happen during next summer. The stipend offered by the company is Rs. 75,000 per month.

CANDIDATE PROFILE:

The most important thing is **resume**. A lot of templates are available on websites with the help of which you can develop one. I used Canva to build my resume. It can be divided into the following sections – personal details, educational qualification, projects (a short description of each – what it is and technologies used), skills, co-curricular activities and certifications/achievements (if any). You can highlight your best projects if you have worked on many. It is advisable to fit your resume to one page and add your photo too. Also, you should be completely aware of what you add in your resume because interviewers will ask a lot of questions based on it. So **do not add anything that you don't know**.

PREPARATION STRATEGY:

So, from my 3rd semester I started learning about STL in C++ and solving basic problems on leetcode and gfg. ACM had organised a HEADFIRST programme through which we were shared materials that had questions and resources compiled for basic topics. This helped me a lot. I didn't prepare on a daily basis initially.

During the last 2 months before the internship drive is when I started coding on a daily basis. Again, ACM documents for the intern preparation helped me. Apart from this, I used this set of questions from : <https://neetcode.io/practice> .

- Start from basics. Get familiarised with coding platforms like geeksforgeeks and leetcode.
- If you are comfortable with C++, learn STL first. <https://www.studytonight.com/cpp/stl/>
- Make a schedule on how you want to cover the topics and stick to it. Firstly, cover these topics – Arrays, strings, linked list, graphs (BFS and DFS), Trees, sort and search. Later focus on – DP, backtracking and bit manipulation.
- Towards the end, practice questions on daily basis.
- For DBMS I referred the following links:
<https://www.javatpoint.com/dbms-interview-questions>
<https://www.interviewbit.com/dbms-interview-questions/>

https://www.tutorialspoint.com/dbms/dbms_transaction.htm

- For OS and OOPS: <https://www.geeksforgeeks.org/commonly-asked-operating-systems-interview-questions/>
https://docs.google.com/spreadsheets/d/1j05cmJ5JI8O9S4bldKbw_jd0Y4KMZ_5liwyl8vjEbCw/edit#gid=0
- For Aptitude: <https://www.indiabix.com/>

PRELIMINARY ROUND:

The first round was an Online Assessment which was held on HackerEarth platform. It had 2 sections – MCQ and Coding.

The MCQ section had 10 questions which was a mix of aptitude and questions related to basics of programming. This section had questions from easy to moderate level.

The coding section had 3 questions – one medium and two hard.

- The first question was related to arrays. It was something like given an array and value k, you can increment a number in array maximum k times by 1. Find the smallest repeating number which can be obtained by using the operation.
- The second question was related to block game. All blocks have to be cleared to finish the game. In one move you can clear a chunk of blocks. If there are blocks above the chunk cleared, they move down. We have to find the minimum number of moves.
- The third one was related to elevator. Like the elevator can go up and down n floors with some constraints that it can't stop in a particular floor and many other constraints. We have to find if we can move from one floor to another using the elevator. This was very complex question.

I was able to solve only the first question using brute force method, where all of the given testcases passed except one which exceeded time limit. I was able to pass only 2-3 testcases passed for the second question. 10 students were shortlisted for the interviews.

TECHNICAL ROUND 1:

This was an offline interview at CUIC. This round went for 40-45 mins. Firstly, I was asked to introduce myself and then was asked my favourite subject. The interviewer then asked me how will you detect a cycle in a linked list. I told him my approach orally and then he asked me to give a pseudocode.

The next question was direct TWO-SUM problem. He asked me to tell the approach alone. And then I was given a set of inputs and outputs and had to guess the question. The inputs were 2 arrays and their sum. The output was a pair of elements. I gave a few possible questions to it (but didn't tell the exact question). The actual question was "Given scores of team A and team B for each match and the total score of the team, find if we can swap any 2 scores to make the total scores of both teams equal". I came up with brute force approach and later optimised it. He then asked me to write the code for it.

I was asked a few questions about MongoDB which I had mentioned in my resume. Finally, I was asked to solve a puzzle. "Given two ropes of same length, material and both burn at the same rate.

Each rope completely burns off in 1 hr. How will determine the exact 30th minute. The answer I gave was, burn each one from opposite ends and the meeting point will be the 30th minute.

TECHNICAL ROUND 2:

This was a very interesting and different round. It was more of a problem-solving round.

The first question was to check if the array contains consecutive numbers. I came up with a basic approach and then the interviewer asked me to come up with other approaches too. He would add in constraints each time like the numbers might not start from 0, the numbers might be negative or repeating. So, I gave him 5-6 possible approaches and he was satisfied. He asked me write the code for some of the approaches.

The next question was given an array of 0s and 1s in sorted order find the first occurrence of 1. I gave linear search and binary search approach. Later, he modified the question to a string of 0s and 1s and I gave him 2 more approaches (bit manipulation).

I was asked a few questions about my resume. He asked me to explain my favourite project and later few more questions on it. I was also asked the difference between MongoDB and SQL(since I mentioned in my resume about MongoDB). Later he questioned me on another project of mine. The round lasted around 40-45 mins.

HR ROUND:

This was the final round and it went for 15-20 mins. I was asked the following questions

- Tell me something about yourself
- Where do you see yourself in 5 years?
- What are your interests and goals.
- Why do you wish to join Chronus? (Make sure you learn about the company before appearing for the interview)
- Feedback of my performance and takeaways from the previous rounds.
- 2-3 questions from my resume
- Biggest setback in college and what have you learnt from it.

TIPS:

- Stay calm and composed. Do not get tensed.
- Know everything that you've put in your resume. Make sure you know everything about your projects.
- Have a good knowledge about DBMS, OOPS and OS.
- Make sure to communicate with interviewer. Do not remain silent. Feel free to tell on how you are thinking to approach the problem.
- Prepare some standard HR questions.
- At the end of interview, you will be asked to if you have any questions for them. Do not say "no". Ask them a few questions like – what kind of projects will I get to work, the work culture and something like that.

Feel free to reach me if you need any help or have more questions at 7299955218. ALL THE BEST!